

Nikunj Pradhan

nikunj.pradhan31@gmail.com | (314)-719-9396 | St. Peters, MO | nikunjpradhan.com

EDUCATION

Missouri University of Science and Technology

Bachelor of Science, Computer Science

GPA: 4.0/4.0

Aug 2023 - Dec 2026

EXPERIENCE

Data Science Co-Op

Jan 2025 – Aug 2025

Hunter Engineering Company

Bridgeton, MO

- Elevated deep learning model workflow accuracy from 56% to 84% through hyperparameter optimization, active learning, and selective data clustering.
- Built a CNN-based regression model predicting 4-point image coordinates using RMSE and MAE loss, improving geometric correction accuracy by 23% and streamlining preprocessing.
- Created and maintained custom ETL pipelines to process metadata and image data from 600+ worksites, ingesting over 2M images into Microsoft SQL Server and AWS S3.
- Automated cross-cloud benchmarking pipelines across AWS and Azure, enabling direct performance comparisons of vendor AI & ML models.
- Reduced model training time by 66% by migrating YOLO detection/classification models to PyTorch Lightning and enabling multi-GPU training with CUDA acceleration.
- Delivered full-stack apps, production model dashboards, dataset analytical applications, and AI-assisted annotation.

Backend Engineer

June 2025 – October 2025

Tally Receipts

St. Louis, MO

- Redesigned and migrated backend from MongoDB to AWS DynamoDB, redesigning schemas for efficient querying, scalability, and data lifecycle management.
- Built LLM-powered receipt categorization service, reducing manual entry effort for users and integrating seamlessly with calendar workflows (Google, Microsoft).
- Refactored services and integrated AWS Lambda, Amplify, and IAM, streamlining backend logic and infrastructure.

Computer Vision Researcher

Aug 2023 – Dec 2024

Missouri University of Science and Technology

Rolla, MO

- Optimized algorithms, increasing blob analysis accuracy 62% through hyperparameter tuning, and algorithmic refinement using C++ OpenCV and MATLAB.
- Designed a C++/MATLAB application simulating 3D triaxial shear tests with deep learning and parallel computing for soil compression modeling with Point Cloud Library.
- Utilized advanced computer vision techniques, including watershed segmentation and 3D model triangulation via ray tracing, and used a KNN-based algorithm for point classification.

PROJECTS

Nepali Devanagari OCR Pipeline | *PyTorch, Python, FastAPI, NLTK, Docker, ONNX*

- Designed a CRNN-based Nepali OCR pipeline covering 130+ characters, combining CNN backbones with LSTM/Bi-LSTM sequence modeling and CTC/Attention mechanisms.
- Collected, annotated, and augmented 1000+ images into 500,000+ training samples and utilized the model via a FastAPI inference service for real-time OCR for text recognition and line detection.

20 Newsgroups Text Classification | *Python, Scikit-learn, Matplotlib, Pandas*

- Compared Naive Bayes, Linear SVM, and Multinomial Logistic Regression on the 20 Newsgroups dataset (20,000 documents, 20 classes), achieving up to 95.3% accuracy, with TF-IDF features and visualizations of confusion matrices.

Movie Recommendation & Watchlist Platform | *FastAPI, React, PostgreSQL, MinIO, Scikit-learn, K3S, Docker*

- Deployed and Developed a full-stack movie watchlist app with personalized recommendations, combining content-based and ALS collaborative filtering, with per-user caching for fast, accurate suggestions.

TECHNICAL SKILLS

Programming Languages: Python, C++, JavaScript, SQL, MATLAB, HTML/CSS

Data Science Tools: Pytorch, Scikit-learn, NLTK, Pandas, NumPy, SciPy, Matplotlib, OpenCV

Databases: PostgreSQL, Microsoft SQL Server, DynamoDB, Firebase, MinIO, Mongoose

Web Development & APIs: FastAPI, React, Node.js, Express.js, WebSockets

Cloud & DevOps: AWS (S3, EC2, Lambda), Docker, Kubernetes, Git, CI/CD, Linux/Ubuntu, Jupyter